

Gateworks Project Audit

Prepared for future faster updates. Read-only audit: no source changes, commits, or pushes were made.

Scope: current stack, GitHub/Vercel connection, Supabase connection, development bottlenecks, recommended workflow, and a fast update process.

1. Current Tech Stack

- Framework: Next.js 15.3.0 in package.json; installed build output reports Next.js 15.5.16.
- Language: TypeScript with strict mode enabled.
- UI: React 19.
- Styling system: Tailwind CSS 3.4, PostCSS, app/globals.css, and custom Tailwind colors in tailwind.config.ts.
- State: Zustand with persisted local stores.
- Database: Supabase Postgres.
- Auth: Supabase Auth is wired. Admin API checks bearer tokens and admin_profiles roles.
- Storage: Supabase Storage. Verified bucket: customer-drawings.
- Hosting: Vercel.
- Deployment: Vercel GitHub integration from dirtcollins/Gateworks.
- Environment variables: .env.local exists and is ignored; .env.example lists Supabase/admin vars. Local env also has Turnstile vars.
- Package manager: npm, based on package-lock.json and scripts. Note: npm was not available in the audit shell PATH, so checks were run with direct local Node package binaries.

Build and Verification Commands

```
npm run dev
npm run typecheck
npm run lint
npm run build
npm run supabase:deploy
npm run supabase:verify
```

There is no real unit or end-to-end test command yet.

2. GitHub / Vercel Connection

- GitHub remote: <https://github.com/dirtcollins/Gateworks.git>
- Production branch: main.
- Current branch: main.
- Remote default branch: origin/main.
- Current commit: 1976584 / 1976584e408c542e9fa2d11ded5e310caf3beb8a.
- Vercel project: gateworks.
- Latest production deployment came from GitHub repo dirtcollins/Gateworks, branch main, commit 1976584.

- Vercel is deploying from GitHub.
- PR previews are likely available through Vercel Git integration, but only production deployments from main were observed in the current deployment list.
- GitHub Actions CI exists locally at .github/workflows/ci.yml, but .github/ is currently untracked, so that CI is not active on GitHub until committed and pushed.

3. Supabase Connection

Remote Supabase access was verified successfully.

Tables

Verified or expected tables include:

```
categories, brands, products, product_variants, product_images, product_reviews, site_users, carts,
cart_items, admin_profiles, admin_audit_logs, admin_settings, companies, company_users, staff_profiles,
addresses, suppliers, supplier_products, inventory_locations, inventory_bins, inventory_items,
inventory_events, orders, order_items, saved_carts, saved_cart_items, customer_saved_jobsites,
customer_drawing_uploads, quotes, quote_items, invoices, invoice_items, payments, purchase_orders,
purchase_order_items, receipts, pick_tickets, pick_ticket_items, deliveries, uploaded_files,
notifications, activity_logs, integration_events, product_demand_events, reorder_rules,
reorder_recommendations, purchasing_notes
```

Main Live Counts Observed

- products: 456
- product_variants: 519
- product_images: 484
- categories: 12
- brands: 11
- admin_settings: 1
- quotes: 4

Storage Buckets

- customer-drawings: public bucket, 50 MB file limit, allowed types include PDF, PNG, JPEG, WEBP, HEIC/HEIF, and octet-stream.

Auth

- Supabase Auth admin API is reachable.
- Admin authorization is implemented through Supabase Auth token validation plus admin_profiles.
- admin_profiles currently has 0 rows, so production admin access may require creating admin profiles before staff can use protected admin APIs.

Migrations

- The project uses SQL deployment files in supabase/.
- It is not currently using Supabase CLI migration files under supabase/migrations/.
- The deploy script applies ordered SQL files directly through the Supabase Management API.

Service-Role Exposure

- No committed service-role key value was found in app/source files.
- .env.local contains secrets and is ignored.
- .env.example has blank secret values.
- SUPABASE_SERVICE_ROLE_KEY is referenced in server routes, server helper code, scripts, and GitHub Actions secrets.
- Risk: lib/supabase-admin.ts does not include a server-only guard. It is not currently imported by client components, but adding import "server-only" later would make accidental exposure harder.

4. What Is Slowing Development Down

- Dirty main: the working tree has many modified and untracked files directly on main, making future updates risky and hard to isolate.
- No active committed CI yet: .github/ is untracked, so the CI workflow is not protecting GitHub until committed.
- No tests beyond typecheck/lint/build: no unit, integration, or Playwright tests for checkout, admin, cart, warehouse, or Supabase workflows.
- Large components: features/admin/catalog/catalog-manager.tsx has 1259 lines; features/admin/inventory/inventory-dashboard.tsx has 1214 lines; components/product-page-client.tsx has 1185 lines; components/admin-dashboard.tsx has 1053 lines; features/admin/quotes/quotes-dashboard.tsx has 981 lines.
- Supabase schema is not migration-based. SQL files work, but future schema changes will be harder to review, rollback, and sequence.
- Several workflows still mix durable Supabase persistence with local Zustand/localStorage prototype state.
- Design system exists but is early: components/ui has useful primitives, but many dashboards still carry repeated table/filter/modal/action logic.
- Local command setup is confusing: package manager is npm, but npm was not available in the execution PATH.
- Build is currently fine. Production build passed and generated 427 pages.

Verification Results

- TypeScript: passed.
- ESLint: passed.
- Next build: passed.
- Supabase remote verify: passed.

5. Recommended Workflow Going Forward

Best Branch Workflow

- Keep main production-only.
- Create a branch per change: codex/<short-task-name>.
- Open a pull request for every production change.
- Let Vercel create preview deployments from branches and PRs.
- Merge to main only after checks and preview review pass.
- Avoid doing new work directly on main.

Best Codex Task Size

- Good task examples: Add saved cart rename button; fix admin inventory quantity validation; split product gallery from product page.
- Too-large examples: Finish admin; make checkout production-ready; improve design.
- Best size is one route, one workflow, or one component family at a time.

Best Checks

- For TypeScript/UI changes: npm run typecheck, npm run lint, npm run build.
- For Supabase/schema/data-access changes: the above plus npm run supabase:verify.
- For checkout/admin/warehouse flows: add browser verification once Playwright or a simple e2e script exists.
- For pricing/inventory/order logic: add focused unit tests before expanding behavior.

Best Release Process

- 1 Start from clean main.
- 2 Create codex/<task>.
- 3 Codex makes a small scoped change.
- 4 Run typecheck, lint, and build.
- 5 Push branch.
- 6 Review Vercel preview.
- 7 Merge PR to main.
- 8 Vercel deploys production.
- 9 If schema changed, apply Supabase migration/deploy intentionally and verify.

What Should Be Automated

- GitHub Actions on PR: npm ci, npm run typecheck, npm run lint, npm run build.
- Optional but valuable: npm run supabase:verify when Supabase env secrets are available.
- Secret scanning.
- Playwright smoke test for home, product page, cart, checkout, and admin auth boundary.
- Vercel preview checks linked to PRs.

What To Add To AGENTS.md

- Require new branches for implementation unless the user explicitly says to work on current branch.
- Require reporting dirty worktree before edits.
- Require lint in the standard check list, not just typecheck/build.
- Require server-only for service-role Supabase modules.
- Require Supabase schema changes to become real migrations, or explicitly document when direct SQL deploy files are being used.
- Require browser smoke checks for UI/routing changes once available.
- Require no durable commerce/admin workflow to rely only on localStorage unless explicitly marked prototype.

Codex Skills

- Custom Codex Skills are not needed yet.
- Better AGENTS.md instructions plus smaller tasks will get most of the speed improvement.
- A custom skill may become useful later for a Gateworks release workflow once the branch/PR/Supabase migration process is stable.

6. Fast Update Workflow

What You Tell Codex

Create a new branch for this task. Make only this change: <specific change>. Do not touch unrelated files. Run typecheck, lint, build, and any Supabase verification needed. Do not commit or push until I approve.

What Codex Changes

- Only the files needed for that feature or fix.
- Existing UI primitives first.
- Existing Supabase helpers and server routes first.
- No broad refactors unless the task is specifically a refactor.

What Codex Checks

- Always: npm run typecheck.
- Usually: npm run lint.
- For UI/routing/build-sensitive changes: npm run build.
- For Supabase/schema/data-access changes: npm run supabase:verify.
- For visual changes: run local dev server and inspect the page.

How You Review It

- Read the changed files.
- Open the Vercel preview or local dev URL.
- Test the exact workflow changed.
- Confirm no unrelated files changed.
- Approve commit, push, and PR only after checks pass.

How It Goes Live

- Merge PR into main.
- Vercel deploys production from main.
- Run or confirm Supabase deployment separately when schema changed.
- Run npm run supabase:verify after schema/data deployment.

Bottom Line

The stack is solid and currently builds cleanly. The main speed problem is workflow discipline: too much uncommitted work on main, no committed CI protection yet, large components, and schema files that are not formal migrations. Clean branch-per-task PRs plus automated checks will make future updates much faster and safer.